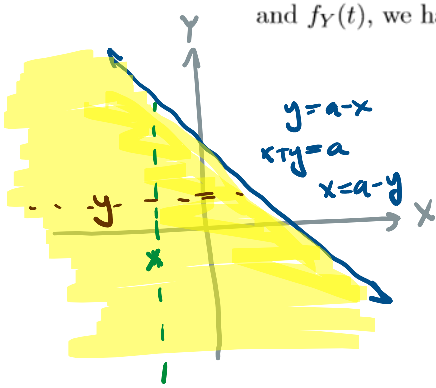
6.3: Sums of Independent Random Variables

Remember if you have two independent continuous random variables X and Y with density $f_X(t)$ and $f_Y(t)$, we have $P(X+Y<a) = \int_{-\infty}^{\infty}\int_{-\infty}^{a-y} f_X(t) f_Y(y)\,dt\,dy$



$-\infty < x < \infty \qquad -\infty < y < a-x$

$-\infty < y < \infty \qquad -\infty < x < a-y$

$f_{XY}(x,y) = f_X(x)\cdot f_Y(y)$

This is a convolution integral: $f(x)\star g(x) := \int_{-\infty}^{\infty} f(t)g(x-t)\,dt$

We can simplify the above as $f_{X+Y} = f_X \star f_Y$

$$f_{X+Y}(a) = \frac{d}{da}\left[P(X+Y\le a)\right] = \frac{d}{da}\left[\int_{-\infty}^{\infty}\int_{-\infty}^{a-y} f_X(t) f_Y(y)\,dt\,dy\right]$$

$$= \int_{-\infty}^{\infty} f_Y(y)\cdot \frac{d}{da}\left[\int_{-\infty}^{a-y} f_X(t)\,dt\right] dy$$

$$= \int_{-\infty}^{\infty} f_Y(y)\cdot f_X(a-y)\,dy = (f_Y \star f_X)(a)$$

Similarly for discrete random variables we have $P(X+Y=n) = \sum_k p_X(k)p_Y(n-k)$

$$P(X+Y=n) = P\left(\bigcup_k \{X=k, Y=n-k\}\right)$$

$$= \sum_k P(X=k, Y=n-k)$$

$$= \sum_k P(X=k)\cdot P(Y=n-k)$$

$$= \sum_k p_X(k)\cdot p_Y(n-k)$$

"convolution sum"

Example: Let $X = Pois(\lambda)$ and $Y = Pois(\gamma)$ be independent random variables.

Prove $X + Y$ is Poisson.

$X = Pois(\lambda) \longrightarrow P_X(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$ FOR $k = 0, 1, \ldots$

$Y = Pois(\gamma) \longrightarrow P_Y(j) = e^{-\gamma} \cdot \frac{\gamma^j}{j!}$ FOR $j = 0, 1, \ldots$

$$P(X+Y=n) = \sum_k P_X(k) \cdot P_Y(n-k)$$

$$= \sum_{k=0}^{n} e^{-\lambda} \cdot \frac{\lambda^k}{k!} \cdot e^{-\gamma} \cdot \frac{\gamma^{n-k}}{(n-k)!}$$

$$= e^{-(\lambda+\gamma)} \cdot \sum_{k=0}^{n} \frac{1}{k!(n-k)!} \lambda^k \cdot \gamma^{n-k}$$

$$= e^{-(\lambda+\gamma)} \cdot \frac{1}{n!} \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \lambda^k \gamma^{n-k}$$

$$= e^{-(\lambda+\gamma)} \cdot \frac{1}{n!} \underbrace{\sum_{k=0}^{n} \binom{n}{k} \lambda^k \gamma^{n-k}}_{\text{BINOMIAL}} \quad \text{SUM} = (\lambda+\gamma)^n$$

$$= e^{-(\lambda+\gamma)} \cdot \frac{(\lambda+\gamma)^n}{n!} = P(\,Pois(\lambda+\gamma) = n\,)$$

$$\therefore \quad X + Y = Pois(\lambda+\gamma).$$

Example: Let $X = Bin(n,p)$ and $Y = Bin(m,p)$ be independent random variables.

Prove $X + Y = Bin(n+m,p)$

$X = Bin(n,p) \longrightarrow p_X(i) = \binom{n}{i} p^i (1-p)^{n-i} \quad i = 0,1,\dots,n$

$Y = Bin(m,p) \longrightarrow p_Y(j) = \binom{m}{j} p^j (1-p)^{m-j} \quad j = 0,1,\dots,m$

$$P(X+Y=k) = \sum_i p_X(i)\, p_Y(k-i)$$

$$= \sum_{i=0}^{k} \binom{n}{i} p^i (1-p)^{n-i} \binom{m}{k-i} p^{k-i} (1-p)^{m-(k-i)}$$

$$= \sum_{i=0}^{k} \binom{n}{i}\binom{m}{k-i} p^k (1-p)^{n+m-k}$$

$$= p^k (1-p)^{n+m-k} \cdot \sum_{i=0}^{k} \binom{n}{i}\binom{m}{k-i}$$

$$\left(\text{HYPERGEOMETRIC:} \quad \sum_{i=0}^{k} \frac{\binom{n}{i}\binom{m}{k-i}}{\binom{n+m}{k}} = 1 \right)$$

$$= p^k \cdot (1-p)^{n+m-k} \cdot \binom{n+m}{k} = P(Bin(n+m,p) = k)$$

So $X+Y = Bin(n+m,p)$ ■

Example: Let X_1, X_2 be independent copies of the uniform random variable $U[0,1]$

Find the density, cumulative distribution function, and expected value of $Y = X_1 + X_2$

$$X_1 = \text{UNIF}([0,1]) \longrightarrow f_{X_1}(t) = \frac{1}{1-0} \quad \text{For } 0 \le t \le 1$$

$$f_{X_1}(t) = 1_{[0,1]}(t) = \begin{cases} 1 & 0 \le t \le 1 \\ 0 & \text{else} \end{cases}$$

$$X_2 = \text{UNIF}([0,1]) \longrightarrow f_{X_2}(t) = 1_{[0,1]}(t)$$

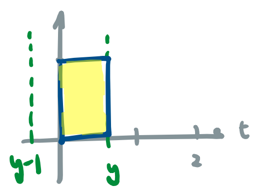
$$f_Y(y) = \int_{-\infty}^{\infty} f_{X_1}(t) f_{X_2}(y-t)\,dt = \int_{-\infty}^{\infty} 1_{[0,1]}(t) \cdot 1_{[0,1]}(y-t)\,dt$$

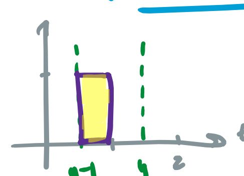
$$= \int_0^1 1_{[0,1]}(y-t)\,dt = \int_0^1 1_{[y-1,y]}(t)\,dt = \boxed{\begin{cases} y & 0 \le y \le 1 \\ 2-y & 1 \le y \le 2 \end{cases}}$$

$0 \le y-t \le 1$

$-y \le -t \le 1-y$

$y \ge t \ge y-1$





$$F_Y(t) = \int_{-\infty}^{t} f_Y(y)\,dy = \int_0^t f_Y(y)\,dy = \boxed{\begin{cases} \frac{1}{2}t^2 & 0 \le t \le 1 \\ 1 - \frac{(2-t)^2}{2} & 1 \le t \le 2 \end{cases}}$$

$$\int_0^t y\,dy = \frac{1}{2}y^2\Big|_0^t = \frac{1}{2}t^2$$

$t \ge 1$:

$$\int_0^t f_Y(t)\,dy = \int_0^1 y\,dy + \int_1^t (2-y)\,dy = \frac{1}{2}y^2\Big|_0^1 + \frac{(2-y)^2}{-2}\Big|_1^t = \left(\frac{1}{2} - 0\right) + \left(\frac{(2-t)^2}{-2} + \frac{1}{2}\right)$$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y)\,dy = \int_0^1 y \cdot y\,dy + \int_1^2 y \cdot (2-y)\,dy = \int_0^1 y^2\,dy + \int_1^2 2y - y^2\,dy$$

$$= \frac{1}{3}y^3\Big|_0^1 + \left(y^2 - \frac{1}{3}y^3\right)\Big|_1^2 = \left(\frac{1}{3} - 0\right) + \left(4 - \frac{8}{3}\right) - \left(1 - \frac{1}{3}\right) = \boxed{1}$$

Example: Let $X_1, X_2, \ldots$ be independent copies of the uniform random variable $U[0,1]$

Define $Y_n := \sum_{k=1}^{n} X_k$, what is $P(Y_n \geq 1)$, and if $N = \min\{n : Y_n \geq 1\}$, what is $E[N]$?

$Y_1 = X_1 \longrightarrow f_{Y_1}(t) = 1 \quad \text{For } 0 \leq t \leq 1$

$Y_2 = X_1 + X_2 \longrightarrow f_{Y_2}(t) = t \quad \text{For } 0 \leq t \leq 1$

$\vdots$

$Y_n = \sum_{k=1}^{n} X_k \longrightarrow f_{Y_n}(t) = \frac{t^{n-1}}{(n-1)!} \quad \text{For } 0 \leq t \leq 1$

$Y_n = Y_{n-1} + X_n$

$f_{Y_n}(t) = f_{Y_{n-1}} * f_{X_n}$

$= \int_{-\infty}^{\infty} \underbrace{f_{Y_{n-1}}(y)}_{\geq 0} \cdot \underbrace{\mathbb{1}_{[0,1]}(t-y)}_{t-1 \leq y \leq t}\, dy$

$= \int_{0}^{t} f_{Y_{n-1}}(y)\, dy$

$P(Y_n \geq 1) = 1 - P(Y_n < 1) = 1 - \int_0^1 f_{Y_n}(t)\,dt = 1 - \int_0^1 \frac{t^{n-1}}{(n-1)!}\,dt = 1 - \left(\frac{t^n}{n!}\right)\Big|_0^1 = \boxed{1 - \frac{1}{n!}}$

$P(N=n) = P(Y_n \geq 1, Y_{n-1} < 1) = P(Y_n \geq 1) - P(Y_{n-1} \geq 1)$

$= \left(1 - \frac{1}{n!}\right) - \left(1 - \frac{1}{(n-1)!}\right) = \frac{1}{(n-1)!} - \frac{1}{n!} = \frac{n-1}{n!}$

$n = 2, 3, \ldots$

$E[N] = \sum_n n \cdot P(N=n) = \sum_{n=2}^{\infty} n \cdot \frac{n-1}{n!} = \sum_{n=2}^{\infty} \frac{n-1}{(n-1)!} = \sum_{n=2}^{\infty} \frac{1}{(n-2)!}$

$= \sum_{m=0}^{\infty} \frac{1}{m!} = \sum_{m=0}^{\infty} \frac{1^m}{m!} = e^1 = \boxed{e}$